3-1.

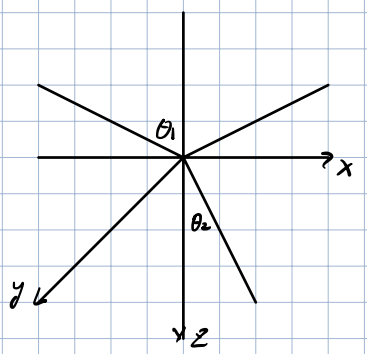
$$n_1\sin\theta_1 = n_2\sin\theta_2 \Rightarrow \frac{1}{2} = 1.7\times\sin\theta_2 \Rightarrow \sin\theta_2 = \frac{1}{2\times1.7} = \frac{1}{3.4}.$$

$$r_S = -\frac{\sin(\theta_1-\theta_2)}{\sin(\theta_1+\theta_2)} \qquad r_P = -\frac{\tan(\theta_1-\theta_2)}{\tan(\theta_1+\theta_2)}$$

$$R_S = |r_S|^2 \qquad R_P = |r_P|^2.$$

3-4.

$$\begin{cases} n\times(E_1+E_1') = n\times E_2. \\ n\times(H_1+H_1') = n\times H_2. \end{cases}$$



分量形式:

$$\begin{cases} (E_{1x}+E_{1x}')\hat{y_0} - (E_{1y}+E_{1y}')\hat{x_0} = E_{2x}\hat{y_0} - E_{2y}\hat{x_0} \\ (H_{1x}+H_{1x}')\hat{y_0} - (H_{1y}+H_{1y}')\hat{x_0} = H_{2x}\hat{y_0} - H_{2y}\hat{x_0} \end{cases}$$

$\Leftrightarrow$

$$\begin{cases} E_{1x}+E_{1x}' = E_{2x} \\ E_{1y}+E_{1y}' = E_{1y} \\ H_{1x}+H_{1x}' = H_{2x} \\ H_{1y}+H_{1y}' = H_{2y} \end{cases}$$

考虑 $H=\sqrt{\frac{\varepsilon}{\mu}}E$.

$$(E_{1p}-E_{1p}')\cos\theta_1 = E_{2p}\cos\theta_2.$$

$$E_{1s}'+E_{1s} = E_{2s}$$

$$a_1(E_{1s}-E_{1s}')\cos\theta_1 = a_2E_{2s}\cos\theta_2$$

$$a_1(E_{1p}'+E_{1p}) = E_{2p}\cdot a_2.$$

$$\therefore \quad \frac{E_{1p}'}{E_{1p}} = r_p = \frac{a_2\cos\theta_1 - a_1\cos\theta_2}{a_2\cos\theta_1 + a_1\cos\theta_2}.$$

$$\frac{E_{1s}'}{E_{1s}} = r_s = \frac{a_1\cos\theta_1 - a_2\cos\theta_2}{a_1\cos\theta_1 + a_2\cos\theta_2}$$

$$\frac{E_{2s}}{E_{1s}} = t_s = \frac{2a_1\cos\theta_1}{a_1\cos\theta_1 + a_2\cos\theta_2}.$$

$$\frac{E_{2p}}{E_{1p}} = t_p = \frac{2a_1\cos\theta_1}{a_1\cos\theta_2 + a_2\cos\theta_1}$$

$$\frac{a_1}{a_2} = \frac{n_1}{n_2} \qquad \theta_1 = \theta_2 = 0.$$

$$\therefore \quad r_s = -r_p = \frac{n_1 - n_2}{n_1 + n_2} \qquad t_s = t_p = \frac{2n_1}{n_1 + n_2}$$

(2) $R_s = |r_s|^2 \qquad R_p = |r_p|^2$

$$T_s = \frac{n_2}{n_1}|t_s|^2 \qquad T_p = \frac{n_2}{n_1}|t_p|^2.$$

3-8. 菲涅尔公式

$$r_s = \frac{\sin(\theta_1-\theta_2)}{\sin(\theta_1+\theta_2)}$$

$$r_p = \frac{\tan(\theta_1-\theta_2)}{\tan(\theta_1+\theta_2)}$$

$$t_s = \frac{2n_1\cos\theta_1}{n_1\cos\theta_1+n_2\cos\theta_2}$$

$$t_p = \frac{2n_1\cos\theta_1}{n_1\cos\theta_2+n_2\cos\theta_1}$$

3-10. (1) $\tilde{n} = n(1+ik) = \sqrt{\frac{\varepsilon}{\varepsilon_0}} = \sqrt{\tilde{\varepsilon}_r}$

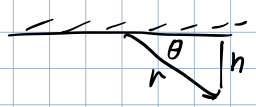
$\tilde{n} = 1.43(1+i0.147)$.

$\therefore$ $n = 1.43$ $K = 0.147$.

↓

实部代折射定律 $\longrightarrow$ θ_2.

(3).



$E = E_0 \exp[i(\tilde{k}\cdot r - \omega t)]$

$= E_0 \exp[i(\tilde{k}\cdot \frac{h}{\cos\theta} - \omega t)]$

$= E_0 \exp[i(n(1+ik)\frac{h}{\cos\theta} - \omega t)]$

$= E_0 \exp(-K\frac{hn}{\cos\theta}) \exp[i(nr - \omega t)]$

$|\frac{E}{E_0}| = e^{-nKh/\cos\theta} = e^{-1}$.

$h = \frac{\cos\theta}{nK} =$

3-10

$\tilde{n}^2 = \tilde{\varepsilon_r} = \frac{\tilde{\varepsilon}}{\varepsilon_0}$ $\quad \tilde{n} = 1.43\,(1 + i0.147)$.

$n = 1.43$. $\quad K = 0.147$

折射角由 n 实部求